

MONETARY LOAN ELIGIBILITY PREDICTION USING LOGISTIC REGRESSION ALGORITHM

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Abstract— In the banking sector, predicting customer loan eligibility is a crucial task. This paper presents a study of using machine learning algorithms to predict loan eligibility of customers based on credit history, income, and others. The algorithms used in the study are Logistic Regression. Using machine learning algorithms the results are demonstrated the potential of in the banking sector to improve loan decision-making processes and increase efficiency. The study utilized a dataset containing information on various loan applications and their respective approvals. The dataset was pre-processed and various machine learning algorithms were applied to the data to determine the best model for predicting loan approval. The findings of this study can be useful for financial institutions in making informed decisions about loan approvals, reducing the risk of default and improving the overall loan approval process. The model is trained using a dataset containing various demographic and financial parameters of loan applicants. The Logistic regression algorithm is utilized to build a predictive model that determines the eligibility of a loan applicant based on the input features. The system aims to modernize the loan approval process by using predictive algorithms to analyze and evaluate loan applications. The study compares different machine learning algorithms, including logistic regression and support vector machines, and evaluates their performance in terms of accuracy and efficiency. The findings also suggest that the use of machine learning can help banks streamline their loan approval processes.

Keywords—predicting, machine learning, support vector machines, logistic regression algorithm

I. INTRODUCTION

The financial industry is constantly evolving, and loan approval processes are no exception. The traditional approach of manual evaluation of loan applications is becoming increasingly obsolete due to the high risk of errors, and the increasing number of loan applications being processed. To address this issue, machine learning algorithms have been developed to automate the loan approval process and improve the overall accuracy of loan approvals.

The objective of this work is to develop a loan eligibility prediction model using the logistic regression algorithm. The model will be trained on a dataset of historical loan applications, and will be able to accurately predict whether a loan applicant is eligible for a loan or not. The work aims to improve the accuracy of loan approvals, reduce the risk of fraud, and streamline the loan approval process.

II. RELATED METHODS

In this paper [1], an important problem in the banking sector, which is assessing a customer's eligibility for a loan. Utilizes machine learning algorithms to predict loan eligibility, which can potentially increase accuracy and efficiency in the loan approval process. Provides a practical application of machine learning in the banking industry. May offer insights into how banks can better manage risk by identifying potentially risky borrowers. The study may be limited in scope and may not be generalizable to other banks or financial institutions. The accuracy of machine learning algorithms may be affected by biases in the data used to train them. There may be privacy concerns related to the use of customer data to train the machine learning models.

In this paper [2], the study is important because loan defaulters can significantly impact a bank's profits or losses, and predicting them can help reduce Non-Performing Assets (NPA). The paper compares different methods for predicting loan defaulters, with a focus on the Logistic Regression model. Personal attributes of customers such as age, purpose, credit history, credit amount, credit duration, etc. are taken into account along with checking account information to calculate the probability of default on loan correctly. By using logistic regression, the paper concludes that banks should not only target rich customers but also assess other attributes that play a vital part in credit granting decisions and predicting loan defaulters. The paper does not mention the size and representativeness of the sample data used for the study. The paper does not compare the Logistic Regression model with other machine learning algorithms for predicting loan defaulters. The paper does not discuss the ethical considerations and potential biases associated with using personal attributes of customers in credit granting decisions.

The paper does not mention the practical implementation of the results and how the findings can be translated into actions to reduce loan default rates in a bank. The model has a high accuracy of 95% in predicting loan eligibility, indicating that it is a reliable and effective tool for loan approval. Financial institutions can use this approach to streamline their loan approval process, saving time and effort while still ensuring accurate predictions. By accurately predicting loan eligibility, financial institutions can better manage credit risk and reduce the likelihood of default. Biased training data: The accuracy of the model is dependent on the quality and representativeness of the

training data. If the training data is biased or incomplete, the model may not be reliable. The model may be overfitted to the training data, meaning that it may not perform well on new data or in different situations. The Random Forest model may not be easily interpretable, making it difficult to understand the underlying factors that contribute to the model's predictions [3].

By automating the loan prediction process, financial institutions can make faster decisions and improve their customer service. AI/ML technologies can help financial institutions reduce the cost of credit operations by automating processes that were previously manual. Financial institutions may become overly reliant on AI/ML models and overlook important factors that cannot be easily quantified, such as a borrower's character or personal circumstances. This can lead to poor decision-making and potentially harmful outcomes for both the institution and the borrower. Financial institutions need to ensure that the data they collect and use for AI/ML models is secure and protected from unauthorized access or misuse [4].

This can help banks make more informed decisions about credit risk and improve their profitability. By using machine learning models to analyze past data, banks can identify potential risks and reduce the likelihood of granting loans to individuals who may not be able to repay them. By automating the credit risk assessment process, banks can make faster decisions and improve their customer service. Machine learning models can sometimes be biased based on historical data, leading to discrimination against certain groups of people. This can result in unfair credit decisions and outcomes. Banks may become overly reliant on machine learning models and overlook important factors that cannot be easily quantified, such as a borrower's character or personal circumstances. This can lead to poor decision-making and potentially harmful outcomes for both the bank and the borrower [5].

By automating the loan approval process, banks can make faster decisions and improve their customer service. Machine learning algorithms can identify potential risks associated with a particular loan application, enabling banks to take appropriate actions to mitigate these risks. The statistical significance of the study results shows that the algorithms have a high degree of accuracy, precision, and recall, making it more likely that the predictions will be correct. The study used a relatively small sample size of only 20 for each of the algorithms, which may not be representative of the broader population of car loan applications. Banks may become overly reliant on machine learning algorithms and overlook other important factors that can influence the loan approval process, such as the borrower's character or personal circumstances. The effectiveness of the algorithms may not be generalizable to other datasets or populations, as different data sets may have different distributions of variables and risk factors [6].

By using classification models to predict loan default, banks offering alternative repayment options. This can reduce the number of defaults and minimize the losses suffered by the bank. By proactively reminding customers to pay their debt, banks can improve their customer service and reputation. By using classification models to predict loan default, banks can allocate their resources more efficiently

and focus on customers who are most likely to default. Classification models can only predict loan default based on historical data and other quantitative factors. This may not take into account qualitative factors such as the borrower's character or personal circumstances, which can also impact their ability to repay the loan. Classification models can be biased based on historical data and other factors, leading to discrimination against certain groups of people. This can result in unfair loan approvals and rejections. Classification models can be complex and difficult to understand, making it hard for regulators and consumers to know how decisions are being made. This lack of transparency can erode trust in the banking sector and lead to negative outcomes for both the bank and the borrower [7].

III. PROPOSED METHOD

The first step in this work is to collect a comprehensive dataset that includes all the relevant features required for loan eligibility prediction. The dataset should be representative of the target population and of good quality. Next, we need to pre-process the data to handle missing values, remove duplicates and irrelevant columns, and encode categorical variables. We also need to explore the data using visualization techniques and identify any outliers or anomalies. Once the data is pre-processed and explored, we need to prepare the data by splitting it into training and testing sets and balancing the dataset if necessary. We also need to select the most relevant features using techniques like correlation analysis or feature importance from a random forest model. The next step is to build a logistic regression model to predict loan eligibility. The distributed data is loaded into the server on an encrypted state. The collection of the data is performed using homomorphic encryption. The collection algorithm does not depend on whether the data is horizontal or vertical. Then, the server and the active party perform logistic regression model train protocol together.

After building the initial model, we can optimize it by tuning the hyper parameters of the logistic regression model using techniques like grid search or random search. We can also experiment with different feature engineering techniques to create more relevant features for the prediction task. Finally, we need to evaluate the model on the testing set and compare its performance with other models like random forest, decision trees, or support vector machines. We also need to analyze the reasons for any differences in performance and identify areas for improvement. Future work can include experimenting with more advanced modelling techniques like neural networks or gradient boosting, exploring different evaluation metrics, and monitoring the model's performance over time. Smearing the analysis of credit scoring of financial institutions, in deed the loaning. In order ot reduce the degree of threat, the crdit scoring method is implemented. Figure 1 is shows the architecture of proposed work.

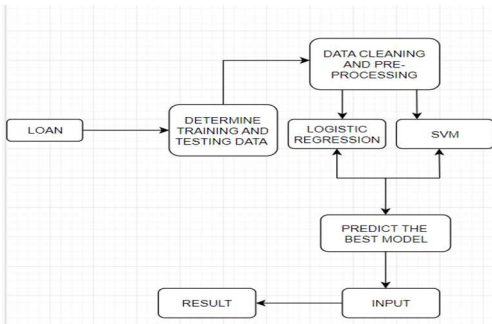


Fig.1. Proposed System Architecture

A. Accuracy Evaluation:

To compare the accuracy of different modules, one typically measures their performance on a set of test data that is distinct from the data used to train the models. This work is intended to use Logistic Regression, Hyper parameter tuning and random forest Classifier to keep track on different accuracies.

B. Loan Eligibility Prediction:

Typically, the model is trained to identify patterns in data that indicate whether or not a person is likely to be approved for a loan. Once trained, the model can be used to predict whether or not a new loan application will be approved. To accomplish this, loan application data is fed into the model, which then generates a prediction based on previously learned patterns in historical data.

IV IMPLEMENTAION & TESTING

This section deals with the dataset used in our research work, Equ 1 shows the calculation of accuracy.

$$\text{Accuracy} = (TP_1 + TN_1) / (TP_1 + TN_1 + FP_1 + FN_1) \quad (1)$$

Precision: Based on true prediction the correctness has been measured. It shown in equation 2.

$$\text{Precision} = TP_1 / (TP_1 + FP_1) \quad (2)$$

Recall (Sensitivity) :

Proportion of patients with a disease who test positive. Recall calculation is shown in equation 3.

$$\text{Recall}_1 = TP_1 / (TP_1 + FN_1) \quad (3)$$

Specificity :

Proportion of patients without the disease who test negative. Equation 4 shows the specificity calculation.

$$\text{Specificity}_1 = TN_1 / (TN_1 + FP_1) \quad (4)$$

Model Building

Let us make predict our first model using logistic regression which is predict binary classification output. Table 1 shows the comparison of various algorithms.

TABLE 1: COMPARISONS OF METHODS

Algorithm	Accuracy
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Logistic Regression	0.82
Hyper Parameter Tuning	0.79
Logistic Regression with Cross-Val	0.85
Random Forest	0.86

	precision	recall	f1-score	support
0	0.94	0.45	0.61	66
1	0.82	0.99	0.89	162

accuracy			0.83	228
macro avg	0.88	0.72	0.75	228
weighted avg	0.85	0.83	0.81	228

0.8333333333333334

This Logistic Regression Model accuracy is 83.33%

Figure 2 shows the evaluation metrics.

Fig.2. Evaluation Metrics

	Loan_ID	Gender	Married	Dependents	Education	Self_Employed	ApplicantIncome	CoapplicantIncome	LoanAmount	Loan_Amount_Term
1	LP001002	Male	No	0	Graduate	No	8448	0	360	
2	LP001003	Male	Yes	1	Graduate	No	4553	1008	126	360
3	LP001006	Male	Yes	0	Graduate	Yes	2845	0	96	360
4	LP001008	Male	No	0	Not Graduate	No	2363	2358	126	360
5	LP001008	Male	No	0	Graduate	No	8880	0	141	360
6	LP001011	Male	No	2	Graduate	No	6447	0	267	360
7	LP001013	Male	No	0	Not Graduate	No	2333	1516	95	360
8	LP001014	Male	Yes	3+	Graduate	No	3636	2054	158	360
9	LP001016	Male	No	2	Graduate	No	4006	1026	160	360
10	LP001020	Male	Yes	1	Graduate	No	12641	10868	348	360
11	LP001024	Male	No	2	Graduate	No	3239	798	79	360
12	LP001027	Male	No	2	Graduate	No	2959	1840	109	360
13	LP001028	Male	Yes	2	Graduate	No	3673	8108	208	360
14	LP001029	Male	No	0	Graduate	No	1933	2848	114	360
15	LP001030	Male	No	2	Graduate	No	1288	1688	17	120
16	LP001032	Male	No	0	Graduate	No	4650	0	125	360
17	LP001034	Male	No	1	Not Graduate	No	2546	0	108	240
18	LP001036	Female	No	0	Graduate	No	3519	0	79	360
19	LP001038	Male	No	0	Not Graduate	No	4687	0	133	360
20	LP001041	Male	No	0	Graduate	No	2688	2688	110	360
21	LP001043	Male	No	0	Not Graduate	No	7669	0	154	360
22	LP001046	Male	Yes	1	Graduate	No	5805	5625	310	360
23	LP001047	Male	No	0	Not Graduate	No	2599	5911	116	360
24	LP001050	Male	No	2	Not Graduate	No	3385	1917	112	360
25	LP001052	Male	Yes	1	Graduate	No	3717	2825	151	360
26	LP001054	Male	No	0	Graduate	Yes	5568	0	191	360
27	LP001056	Male	No	0	Graduate	No	2798	2253	122	360
28	LP001073	Male	Yes	2	Not Graduate	No	4235	1648	119	360
29	LP001086	Male	No	0	Not Graduate	No	1442	0	59	360

Fig.3. Training data

	Loan_ID	Gender	Married	Dependents	Education	Self_Employed	ApplicantIncome	CoapplicantIncome	LoanAmount	Loan_Amount_Term
1	LP001018	Male	Yes	0	Graduate	No	3738	0	110	360
2	LP001022	Male	Yes	1	Graduate	No	3075	1008	126	360
3	LP001031	Male	Yes	2	Graduate	No	5008	1008	208	360
4	LP001035	Male	No	2	Graduate	No	2348	2548	100	360
5	LP001037	Male	No	0	Not Graduate	No	3276	0	79	360
6	LP001039	Male	Yes	0	Not Graduate	Yes	2168	3422	162	360
7	LP001045	Female	No	1	Not Graduate	No	2228	0	99	360
8	LP001058	Male	Yes	2	Not Graduate	No	3881	0	147	360
9	LP001059	Male	Yes	2	Graduate	No	19833	0	289	240
10	LP001067	Male	No	0	Not Graduate	No	2649	2649	123	360
11	LP001078	Male	No	0	Not Graduate	No	3891	0	80	360
12	LP001082	Male	Yes	1	Graduate	No	2165	1516	162	360
13	LP001083	Male	No	3+	Graduate	No	4188	0	46	360
14	LP001084	Male	Yes	2	Graduate	No	12173	0	168	360
15	LP001088	Female	No	0	Graduate	No	4688	0	124	360
16	LP001089	Male	No	1	Graduate	No	5867	0	131	360
17	LP001105	Male	No	2	Graduate	No	4553	2916	200	360
18	LP001107	Male	Yes	3+	Graduate	No	2798	123	128	360
19	LP001108	Male	Yes	0	Graduate	No	8228	7916	300	360
20	LP001115	Male	No	0	Graduate	No	1308	3470	100	180
21	LP001121	Male	Yes	1	Not Graduate	No	1688	1626	40	360
22	LP001124	Female	No	3+	Not Graduate	No	2883	0	26	180
23	LP001126	Male	No	0	Graduate	No	3898	0	101	360
24	LP001135	Female	No	0	Not Graduate	No	2768	0	125	360
25	LP001140	Male	Yes	0	Graduate	No	5488	4088	200	360
26	LP001153	Male	No	0	Graduate	No	0	24000	148	360
27	LP001162	Male	Yes	2	Graduate	No	4263	1285	140	360
28	LP001169	Male	Yes	0	Graduate	No	7608	2790	275	360
29	LP001174	Male	No	0	Graduate	No	3772	833	57	360

Fig.4. Test data

Figure 3 and 4 shows the testing and training of data. The research findings of this study demonstrate the potential of using machine learning algorithms, specifically Logistic Regression, to predict loan eligibility of customers in the banking sector. By analysing various factors such as credit history, income, and debt-to-income ratio, the model can make informed decisions about loan approvals and improve the overall loan approval process. The study utilized a dataset of loan applications and their respective approvals, which was pre-processed and analysed using various machine learning algorithms. The results show that the

Logistic Regression algorithm performed well in terms of accuracy and efficiency compared to other machine learning algorithms such as support vector machines. The findings suggest that the use of predictive algorithms can help financial institutions streamline their loan approval processes and reduce the risk of loan defaults. The model trained on the dataset containing demographic and financial parameters of loan applicants could be useful for banks and other financial institutions to make informed decisions about loan approvals, reduce the risk of default, and improve the overall efficiency of the loan approval process. The size of the dataset is 367 unique values and the dataset fetched from kaggle database.

IV CONCLUSION

This work, Monetary Loan Eligibility Prediction using Logistic Regression Algorithm has been successful in achieving its academic objectives. Through experimentation, it was found that random forest outperformed logistic regression in terms of accuracy, and hyper-parameter tuning did not improve the accuracy of the models significantly. Another takeaway was that hyper-parameter tuning may not always lead to improved performance, and thus, it is essential to evaluate the effectiveness of tuning. The success of the approach adopted in the work was demonstrated by the performance of the random forest algorithm in predicting loan eligibility. The accuracy achieved was satisfactory and the model can be used for practical purposes. In terms of planned outputs and performance targets, the work has achieved its objectives in evaluating different Machine Learning algorithms for predicting loan eligibility and identifying the most accurate algorithm. The benefits of the work include the development of a model for predicting loan eligibility and the potential for use in practical applications such as credit scoring.

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